

Brock Allan

brockallan55@gmail.com | 951-655-0133 | [LinkedIn](#) | Moreno Valley, CA, 92551 | [Github](#)

Education

University of California, Irvine

Bachelor of Science in **Computer Science**, GPA 3.6

- **Coursework:** Data Structures & Algorithms, Security & Privacy, Information Retrieval, System Design, Operating Systems, Computational Complexity, Differential Equations, Machine Learning
- **Distinctions:** Campuswide Honors Collegium, Published AI & Game Design Thesis, Dean's Honor List, CLAP certificate, Chancellors Excellence, FFA President, NHS President

Skills & Tools

- **Coding Languages:** Python, Java, JavaScript, C, C++, C#, Lisp, Prolog, HTML, Assembly, R
- **Industry Software:** Blender, Unity, Unreal Engine, Photoshop, Vim, Git, Spreadsheets, GDB

Technical Experience

Full-Stack Cybersecurity Lab & Network Intrusion Suite

March 2026 – June 2026

- Architected and led the implementation of an exploit and network analysis suite, leading a partner to identify **critical security vulnerabilities** across **software, web applications, and network layers**.
- Engineered a network intrusion detector in **Python** utilizing packet parsing libraries to process **multi-gigabyte .pcap** files, successfully flagging unauthorized **SYN port scanners** and **ARP spoofing** attempts from live traffic streams with a **98% success rate**.
- Designed and executed defenses against **control flow hijacking** exploits using **GDB**, that would otherwise bypass **stack-based memory protection** mechanisms such as **ASLR** and **DEP**, overriding return addresses and deploying **shellcode**.

Web Search Engine & Crawler

January 2026 – March 2026

- Architected and Lead development of a web search engine with **multi-threaded** crawler from scratch in **Python**, achieving **sub-300ms** query response times from a dataset of **50,000+ web pages**.
- Advised project members to engineer crawling logic that adheres to **Robots.txt politeness policies** and implemented **frontier-management** strategies to avoid **crawler traps** and **dead-end pages**.
- Spearheaded an optimized scraper to target and extract **high-density textual information**, creating a quality **inverted search index** under strict **operational and memory constraints**.

Custom Operating System

September 2025 – January 2026

- Developed a custom **Unix-like shell for x86/RISC-V in C** including a **recursive parser** that handles multi-stage **pipelines**, **I/O redirection**, and **comment-aware grammar** for automated script execution.
- Created **kernel level system calls** from scratch including **ps**, to monitor process states such as **Running**, **Sleeping**, and **Zombie**.
- Expanded **Exec** to identify **non-ELF text scripts** and automatically re-route execution through the designated **shell interpreter**.

Professional Experience

Learning Assistant, Intro to Programming, Shannon Alfaro, UCI

September 2025 - Current

- Instructed College students in how to code and design programs to achieve **foundational proficiency in core computing concepts**.
- Closely guided two groups of 6 students through technical coursework, hosted regular **office hours** to clarify complex topics and provided **detailed feedback** on assignments.
- Directed the LA team to **streamline communication**, provided **actionable feedback** to course instructors, and served as the **primary student advocate** to drive course improvements.

Computer Consultant & Data Analyst, Freedom Tax Insurance, Dana Burges

May 2020 – June 2024

- Oversaw the maintenance and repair of **mission-critical company infrastructure**, maximizing hardware lifecycle through **rigorous diagnostic testing**.
- Authored comprehensive executive reports, using **R**, that identified **operational bottlenecks** and delivered **data-backed recommendations** to senior leadership.

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Skills & Tools

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- **Industry Software:** Blender, Unity, **Unreal Engine**, Photoshop, **Vim**, **Git**, Spreadsheets, **GDB**
- **Industry Models:** Diffusion Models, SDEdit, UNet Architecture, Bilinear Interpolation

Technical Experience

Limitations of AI & LLMs in Game Design Methodology – Thesis

September 2025 – June 2026

- Led Advanced AI Research & Agile Workflows via acting as the Product Owner under the **Scrum framework**. Led a multi-stage empirical study evaluating the capabilities and structural constraints of state-of-the-art LLMs (including **GPT-4.3** and **GPT-5.5**), managing iterative sprints, analyzing critical LLM anomalies like **context drift** and **token-level optimization**, and adapted pipelines across sequential interactions.
- Designed and evaluated complex natural language programming strategies ranging from **Zero-Shot** and **Few-Shot** to **Simulated Tuning (Chain-of-Hindsight and Self-Correction paradigms)**, to optimize code quality, reduce intrinsic and extrinsic hallucinations, and mitigate downstream algorithmic defects in generative systems.
- Codified a multi-dimensional quantitative rubric mapping 6 distinct variables (Clarity, Flow, Balance, Length, Integration, Fun) onto a **1–7 scale** to empirically benchmark human-AI interaction. Isolated critical performance deltas demonstrating that guided context-aligned prompt engineering raised user experience metrics to a peak score of **6.33 / 7.0**.

Generating New Forms of Informal Expressions

January 2026 - March 2026

- Led a research project that explored the capability of **LLMs** to produce informal or novel expressions, including idiomatic and slang, using **Single-shot, Few-shot, and Fine-Tuning**.
- **Gemma-3-4b-it**, was fine tuned using **Low Rank Adaptation and Cross Entropy Loss**, then evaluated for Uniqueness, Creativity, Parts of speech, and Semantic Fluency.
- Gained hands-on expertise in working with models from **HuggingFace** and tuning those models for use in other scenarios.

Sentiment Analysis of IMDB Movie Reviews

September 2025 - January 2026

- **Classified** movie reviews as either positive or negative based solely on textual content.
- Implemented **Bag-of-words** using bigrams, **Unsupervised** learning through K-means ++ Clustering, and a **Random Forest** with **XGBoost** classifiers.
- The model reached **86.9%** testing accuracy after **hyperparameter optimization** and **5-Fold Cross Validation**.

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The Dark Lord Beckons

July 2025 - Current

- Created a **3D-2D Rogue-Lite RPG** from scratch using self made **Unity** assets and physics.
- Developed original assets, scripts, lighting, **animations**, and the environment in Unity. Scripts and animations were coded in **C#**.
- Developed **functional** game components and iteratively improved game **design**.

Rocky Lagoon

April 2025 - June 2025

- Created an **immersive** VR pirate-themed game inspired by interactive Disneyland rides.
- Developed original assets, scripts, lighting, animations, and models in Unity in **C#**.
- Gained hands-on expertise in producing **high-quality game assets** and publishing polished **VR** games.

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Freelance Game Developer

June 2024 - Current

- Utilized game engines such as **Unity** and **Unreal Engine 5**, programming in **C++** and **C#**, to create immersive games in **VR**, **2D**, **3D**, and **FPS** genres.
- Designed gameplay systems, developed assets, and implemented interactive mechanics from the ground up for projects such as “Rocky Lagoon.”